MATH329 · OPERATIONS RESEARCH

Mock Midterm Solutions

A practice exam in the midterm's format · 60 minutes · 30 marks

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01 Solution Q1: True / False

#	Answer	Reason
1	**T**	This is its definition.
2	**F**	A slack is added to a $\leq$ constraint. A surplus is subtracted from a $\geq$ constraint.
3	**F**	For max the entering variable has the most negative coefficient.
4	**T**	This is the condition for alternative optimal solutions.
5	**T**	This is the feasibility condition.
6	**F**	The right-hand side must be nonnegative.
7	**T**	These are the nonbasic variables.

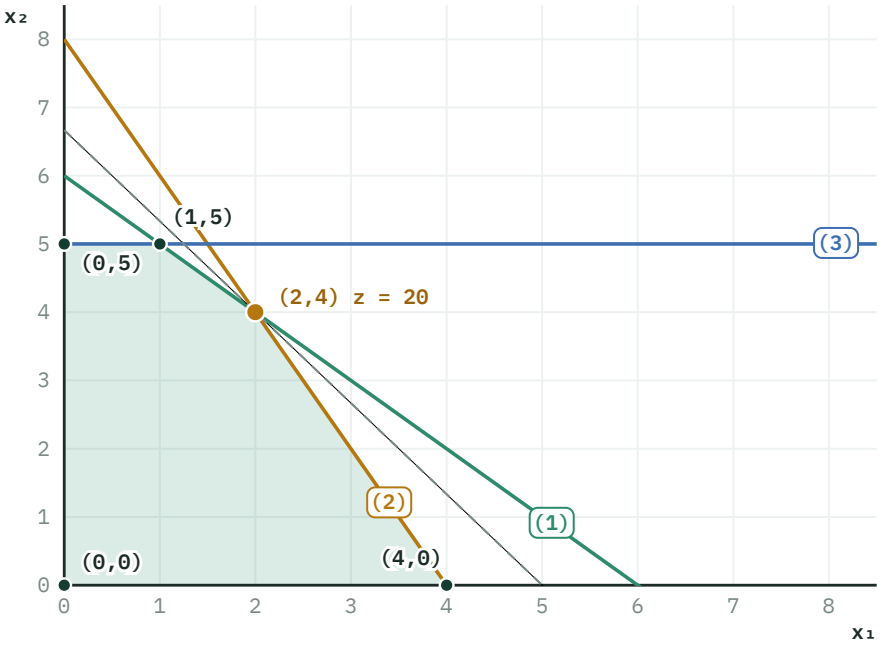
02 Solution Q2: graphical

1 Prepare the constraints

(1) $x_1 + x_2 = 6$ points (0, 6) and (6, 0)

(2) $2x_1 + x_2 = 8$ points (0, 8) and (4, 0)

(3) $-x_2 \geq -5 \times (-1) \rightarrow x_2 \leq 5$ horizontal line at $x_2 = 5$



2 The corner points

Corner	Where it comes from	$Z = 4x_1 + 3x_2$
(0, 0)	origin	0
(4, 0)	(2) with the x_1 axis	16
(2, 4)	(1) with (2)	**20**
(1, 5)	(1) with (3)	19
(0, 5)	(3) with the x_2 axis	15

(2) − (1): $x_1 = 2$ $\quad 2 + x_2 = 6 \quad \rightarrow \quad x_2 = 4$

$Z = 20$ at $x_1 = 2$ and $x_2 = 4$

03 Solution Q3: standard form

1. Graphical method and Algebraic method.

$2x_2 = -5 \quad \times (-1) \quad \rightarrow \quad -2x_2 = 5$

$x_2 = x_2^+ - x_2^-$ in every place x_2 appears

Minimize $Z = 5x_1 - 4x_2^+ + 4x_2^- + 0s_1 + 0s_2$

S.to $3x_1 + x_2^+ - x_2^- + s_1 = 10$

$x_1 - 2x_2^+ + 2x_2^- - s_2 = 4$

$-2x_2^+ + 2x_2^- = 5$

$x_1, x_2^+, x_2^-, s_1, s_2 \geq 0$

04 Solution Q4: standard form and simplex

a Standard form

$$\text{Maximize } Z = 5x_1 + 4x_2 + 0s_1 + 0s_2$$

$$\text{S.to} \quad 6x_1 + 4x_2 + s_1 = 24$$

$$x_1 + 2x_2 + s_2 = 6$$

$$x_1, x_2, s_1, s_2 \geq 0$$

b Simplex

Basic	x_1	x_2	s_1	s_2	Sol.	Ratio
s_1	**6**	4	1	0	24	4
s_2	1	2	0	1	6	6
Z	−5	−4	0	0	0	
x_1	1	$\frac{2}{3}$	$\frac{1}{6}$	0	4	6
s_2	0	$\frac{4}{3}$	$-\frac{1}{6}$	1	2	$\frac{3}{2}$
Z	0	$-\frac{2}{3}$	$\frac{5}{6}$	0	20	
x_1	1	0	$\frac{1}{4}$	$-\frac{1}{2}$	3	
x_2	0	1	$-\frac{1}{8}$	$\frac{3}{4}$	$\frac{3}{2}$	
Z	0	0	$\frac{3}{4}$	$\frac{1}{2}$	21	

Iteration 1: x_1 enters because -5 is the most negative. The ratios are $\frac{24}{6} = 4$ and $\frac{6}{1} = 6$, so s_1 leaves and the pivot is 6.

Iteration 2: $-\frac{2}{3}$ remains under x_2. The ratios are $\frac{4}{\frac{2}{3}} = 6$ and $\frac{2}{\frac{4}{3}} = 1.5$, so s_2 leaves and the pivot is $\frac{4}{3}$.

Optimum $Z = 21$ at $x_1 = 3$ and $x_2 = \frac{3}{2}$

Check: $Z = 5(3) + 4(1.5) = 15 + 6 = 21$

05 Grade yourself

Question	How to mark	Marks
Q1	One mark per item.	7
Q2	One mark per line, one for the objective line, two for the region, one for the optimum point, one for the answer.	8
Q3	One mark for the methods and one for each of the five lines.	6
Q4	Two marks for the standard form, two for the first table, two for each later table, one for the optimum. Each arithmetic slip costs half a mark.	9

The marking here is approximate, based on the past answer keys.